clear in my mind that we should not only be able to make the rocket engine ourselves, we should also be able to upgrade it to launch heavier satellites into the geosynchronous transfer orbit. The Viking engine we were working on (Viking-4) had a thrust of sixty tonnes, and I was dreaming of a 100-plus tonne engine to launch heavy satellites. For many, this was a joke, as India had nothing in hand. We were just tottering around the idea of developing a liquid engine with the French, and here I was dreaming things that made several friends burst into laughter.

Some who pulled my leg did it often to defuse my tension than bring me down. Some others back in India, who either did not yet realise the potential of liquid propulsion or just refused to think beyond solid, tried more silent games. But, having come to Vernon and set up the team on the most ambitious of projects, I could not care less for my detractors. As for those who made my plans the butt of many banter, I laughed along.

Truth be told, ISRO still did not have approval for a liquid engine rocket. Then why was I sent to France with a team to work on liquid propulsion? Dhawan himself laughingly answered this in a meeting, taking some by surprise. 'Nambi was chewing my ears and eating my brain, talking about this liquid thing. I thought I can be at peace by sending him away for a few years.' I joined the others in the laughter, but I knew that Dhawan had realised the potential of liquid propulsion, though ISRO had not put in place a system to pursue the technology.

And this meant that even if my team went back to India with the know-how and the 'know-why', we wouldn't be allowed to put together a rocket. And that is what happened once we returned after mastering the engine in 1979. I was not allowed to fabricate the engine.

But I did not need permission to make a model of the engine, which I had named Vikas. With the enthusiasm of a nursery kid using matchboxes to make a train that he can never have in real, I put together a model, of the same size as the original, with all the moving parts. Only the materials were different and, of course, the engine did not work. But it looked as good as the real one. This was not just an act in self-gratification. I had a plan.